

# Лабораторная работа №3 Планирование локальной сети организации

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Российский университет дружбы народов

Познакомиться с принципами планирования локальной сети организации

# Выполнение лабораторной работы

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# Создание нового проекта.

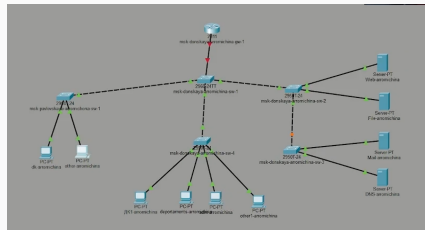
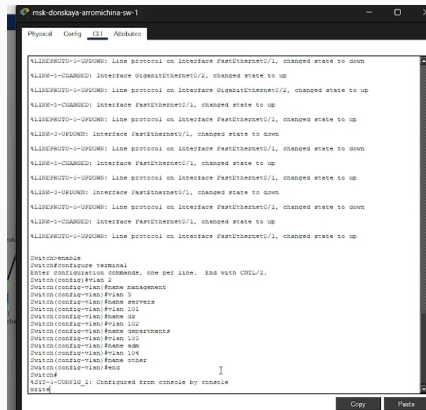


Рис. 1: Повторяем схемы из лабораторной работы

# Настройка msk-donskaya-arromichina-sw-1.



```
msk-donskaya-arromichina-sw-1
Physical Config CLI Attributes

LINEPROTO-3-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down
LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down
LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CTRL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name management
Switch(config-vlan)#vlan 3
Switch(config-vlan)#name servers
Switch(config-vlan)#vlan 101
Switch(config-vlan)#name de
Switch(config-vlan)#vlan 102
Switch(config-vlan)#name departments
Switch(config-vlan)#vlan 103
Switch(config-vlan)#name all
Switch(config-vlan)#vlan 104
Switch(config-vlan)#name other
Switch(config-vlan)#exit
Switch#
MSR2-3-CUIFIQ_1: Configured from console by console
Switch#
```

Рис. 2: Начальная настройка msk-donskaya-arromichina-sw-1

# Настройка msk-donskaya-arromichina-sw-2

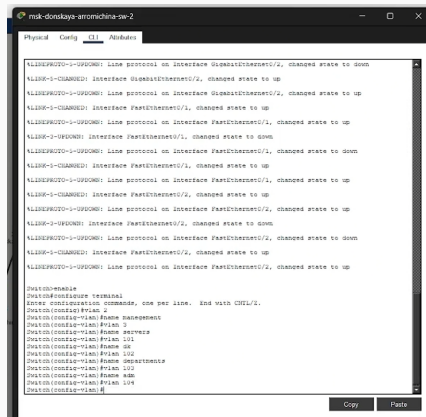
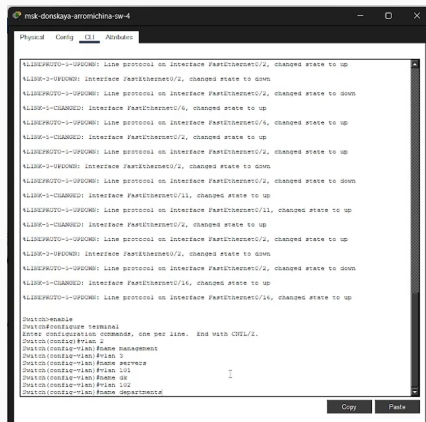


Рис. 3: Начальная настройка msk-donskaya-arromichina-sw-2

# Настройка msk-donskaya-arromichina-sw-4



The screenshot shows a window titled "msk-donskaya-arromichina-sw-4" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying a list of interface status changes and a set of configuration commands.

```

$LINEPROTO-1-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
$LINK-2-UPDOWN: Interface FastEthernet0/2, changed state to down
$LINEPROTO-1-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
$LINK-3-CHANGED: Interface FastEthernet0/6, changed state to up
$LINEPROTO-1-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
$LINK-1-CHANGED: Interface FastEthernet0/2, changed state to up
$LINEPROTO-1-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
$LINK-2-UPDOWN: Interface FastEthernet0/2, changed state to down
$LINEPROTO-1-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
$LINK-3-CHANGED: Interface FastEthernet0/11, changed state to up
$LINEPROTO-1-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up
$LINK-1-CHANGED: Interface FastEthernet0/2, changed state to up
$LINEPROTO-1-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
$LINK-2-UPDOWN: Interface FastEthernet0/2, changed state to down
$LINEPROTO-1-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
$LINK-1-CHANGED: Interface FastEthernet0/16, changed state to up
$LINEPROTO-1-UPDOWN: Line protocol on Interface FastEthernet0/16, changed state to up

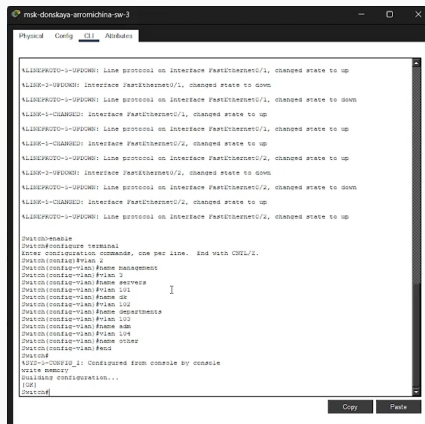
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name management
Switch(config-vlan)#vlan 3
Switch(config-vlan)#name servers
Switch(config-vlan)#vlan 101
Switch(config-vlan)#name dk
Switch(config-vlan)#vlan 102
Switch(config-vlan)#name department#

```

At the bottom right of the CLI window, there are "Copy" and "Paste" buttons.

Рис. 4: Начальная настройка msk-donskaya-arromichina-sw-4

# Настройка msk-donskaya-arromichina-sw-3

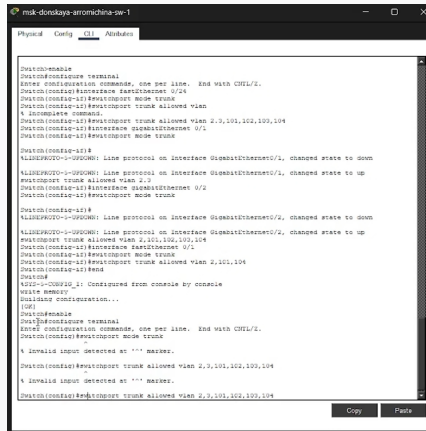


```
msk-donskaya-arromichina-sw-3
Physical Config CLI Attributes

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINK-1-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-1-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-3-UPDOWN: Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINK-1-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration terminal
Enter configuration commands, one per line. End with CTRL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name management
Switch(config-vlan)#vlan 3
Switch(config-vlan)#name servers
Switch(config-vlan)#vlan 101
Switch(config-vlan)#name ge
Switch(config-vlan)#vlan 102
Switch(config-vlan)#name departments
Switch(config-vlan)#vlan 103
Switch(config-vlan)#name adm
Switch(config-vlan)#vlan 104
Switch(config-vlan)#name other
Switch(config-vlan)#end
Switch#
V202-3-COVP0_1: Configured from console by console
Write memory
Validate configuration...
[OK]
Switch#
```

Рис. 5: Начальная настройка msk-donskaya-arromichina-sw-3



```
msk-donskaya-arromichina-sw-1
Physical Config CLI Attributes

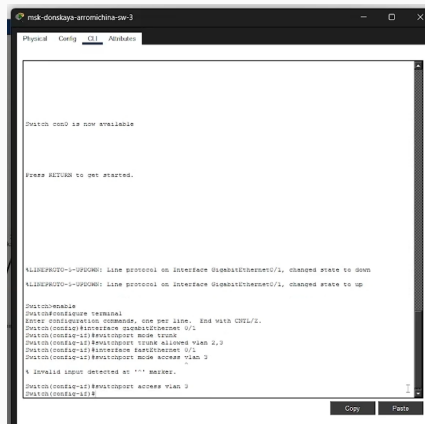
Switch>enable
Switch>configure terminal
Enter configuration commands, one per line. End with CTRL/Z.
Switch(config)#interface fastEthernet 0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan
% Incomplete command.
Switch(config-if)#switchport trunk allowed vlan 2,3,101,102,103,104
Switch(config-if)#interface GigabitEthernet 0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Switch(config-if)#interface fastEthernet 0/2
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
Switch(config-if)#interface fastEthernet 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 2,101,104
Switch(config-if)#end
Switch#
Switch#>copy running-config startup-config
Building configuration...
[OK]
Switch#enable
Switch>configure terminal
Enter configuration commands, one per line. End with CTRL/Z.
Switch(config)#switchport mode trunk
% Invalid input detected at '^' marker.
Switch(config)#switchport trunk allowed vlan 2,3,101,102,103,104
% Invalid input detected at '^' marker.
Switch(config)#switchport trunk allowed vlan 2,3,101,102,103,104
```

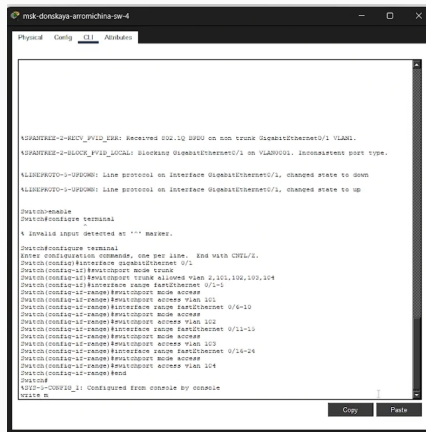
Рис. 6: Настройка msk-donskaya-arromichina-sw-1

## Настройка msk-donskaya-arromichina-sw-3



**Рис. 7:** Начальная настройка msk-donskaya-arromichina-sw-3

# Настройка msk-donskaya-arromichina-sw-4



```
msk-donskaya-arromichina-sw-4
Physical Config CLI Attributes

%SPANFTRM-2-RECV_FVID_ERR: Received 802.1Q BPDU on non trunk GigabitEthernet0/1 VLAN1.
%SPANFTRM-2-BLOCK_FVID_LOCAL: Blocking GigabitEthernet0/1 on VLAN0001. Inconsistent port type.

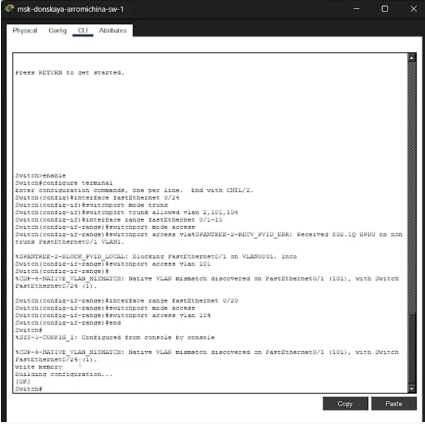
%LINEPROTO-3-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-3-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Switch#enable
Switch#configure terminal
Switch(config)#
% Invalid input detected at '^' marker.

Switch(config)#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface GigabitEthernet 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 2,101,102,103,104
Switch(config-if)#interface range fastEthernet 0/1-1
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 101
Switch(config-if-range)#interface range fastEthernet 0/6-10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 102
Switch(config-if-range)#interface range fastEthernet 0/11-15
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 103
Switch(config-if-range)#interface range fastEthernet 0/16-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 104
Switch(config-if-range)#end
Switch#
%V2-3-0-CONFID_1: Configured from console by console
write m
```

Рис. 8: Настройка msk-donskaya-arromichina-sw-4

# Настройка msk-donskaya-arromichina-sw-1



```
msk-donskaya-arromichina-sw-1
Physical Config CLI Attributes

Press RETURN to get started.

Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface FastEthernet 0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 2,101,104
Switch(config-if)#interface range FastEthernet 0/1-10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlanPANTREE-2-RCV_PVID_ERR: Received PVID 104 on non
trunk FastEthernet0/1 VLAN1.

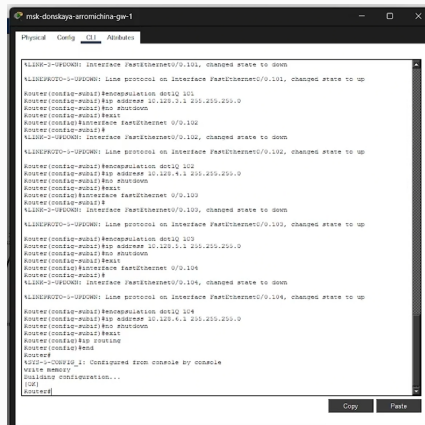
%PANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/1 on VLAN1001. Inco
Switch(config-if-range)#switchport access vlan 101
Switch(config-if-range)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (101), with Switch
FastEthernet0/24 (1).

Switch(config-if-range)#interface range FastEthernet 0/20
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 104
Switch(config-if-range)#end
Switch#
%SYS-3-CONFID_1: Configured from console by console

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (101), with Switch
FastEthernet0/24 (1).
Write memory
Building configuration...
[OK]
Switch#
```

Рис. 9: Настройка msk-donskaya-arromichina-sw-1

# Настройка msk-donskaya-arromichina-gw-1



```
msk-donskaya-arromichina-gw-1
Physical Config CLI Attributes

$LINK-2-UPDOWN: Interface FastEthernet0/0.101, changed state to down
$LINKPROTO-3-UPDOWN: Line protocol on Interface FastEthernet0/0.101, changed state to up
Router(config-subif)#encapsulation dot1Q 101
Router(config-subif)#ip address 10.120.3.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface FastEthernet 0/0.102
Router(config-subif)#
$LINK-2-UPDOWN: Interface FastEthernet0/0.102, changed state to down
$LINKPROTO-3-UPDOWN: Line protocol on Interface FastEthernet0/0.102, changed state to up
Router(config-subif)#encapsulation dot1Q 102
Router(config-subif)#ip address 10.120.4.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface FastEthernet 0/0.103
Router(config-subif)#
$LINK-2-UPDOWN: Interface FastEthernet0/0.103, changed state to down
$LINKPROTO-3-UPDOWN: Line protocol on Interface FastEthernet0/0.103, changed state to up
Router(config-subif)#encapsulation dot1Q 103
Router(config-subif)#ip address 10.120.5.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface FastEthernet 0/0.104
Router(config-subif)#
$LINK-2-UPDOWN: Interface FastEthernet0/0.104, changed state to down
$LINKPROTO-3-UPDOWN: Line protocol on Interface FastEthernet0/0.104, changed state to up
Router(config-subif)#encapsulation dot1Q 104
Router(config-subif)#ip address 10.120.6.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#ip routing
Router(config)#end
Router#
$V2-3-CONFIG_1: Configured from console by console
write memory
Building Configuration...
[OK]
Router#
```

Рис. 10: настройка msk-donskaya-arromichina-gw-1

# Настройка web-arromichina 10.128.0.0.16

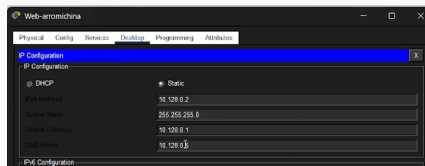
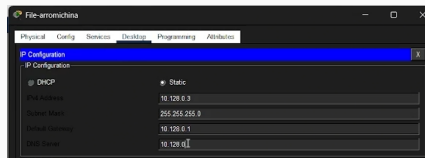


Рис. 11: настройка web-arromichina 10.128.0.0.16

# Настройка file-arromichina 10.128.0.0.16



**Рис. 12:** Настройка file-arromichina 10.128.0.0.16

# Настройка mail-arromichina 10.128.0.0.16

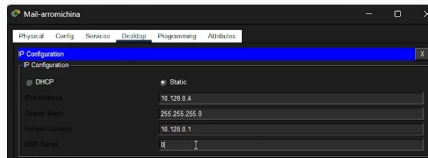


Рис. 13: Настройка mail-arromichina 10.128.0.0.16

# Настройка дк1-arromichina 10.128.0.0.16

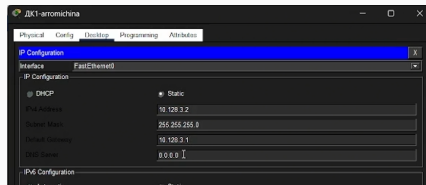


Рис. 14: Настройка дк1-arromichina 10.128.0.0.16

# Настройка dk-arromichina 10.128.0.0.16

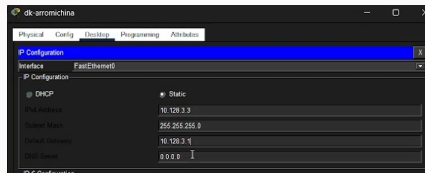


Рис. 15: Присвоение статического IP-адреса и маски подсети.

# Настройка deportaments-arromichina 10.128.0.0.16

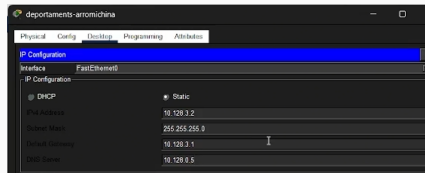


Рис. 16: Настройка deportaments-arromichina 10.128.0.0.16

# Настройка adm-arromichina 10.128.0.0.16.

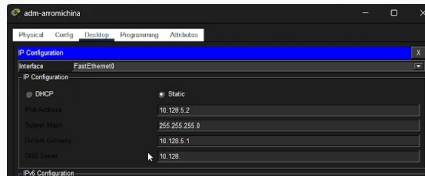


Рис. 17: Настройка adm-arromichina 10.128.0.0.16

# Настройка other-arromichina 10.128.0.0.16

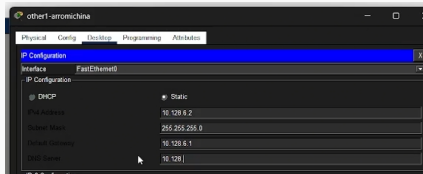


Рис. 18: Настройка other-arromichina 10.128.0.0.16

# Настройка dr-arromichina 10.128.0.0.16

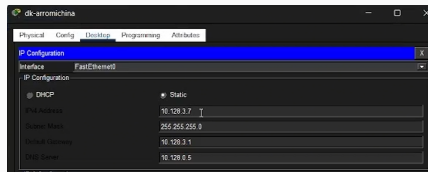


Рис. 19: Настройка dr-arromichina 10.128.0.0.16

# Настройка other-arromichina 10.128.0.0.16

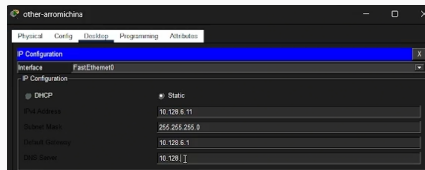


Рис. 20: Настройка other-arromichina 10.128.0.0.16

# Настройка dns-arromichina 10.128.0.0.16

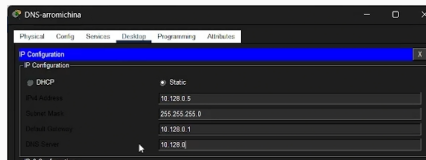


Рис. 21: Настройка dns-arromichina 10.128.0.0.16

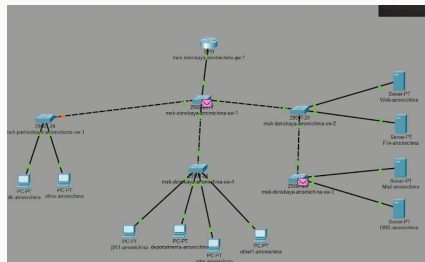
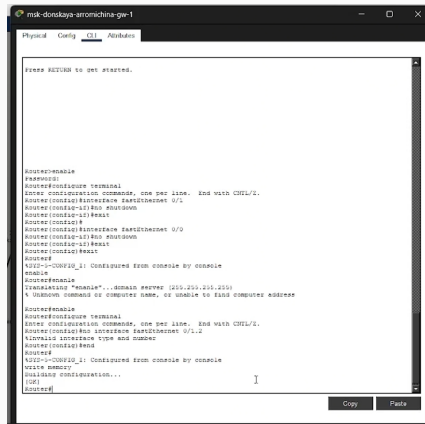


Рис. 22: Настроили все, запускаем на проверку

# настройка msk-donskaya-arromichina-gw-1



The screenshot shows a terminal window titled "msk-donskaya-arromichina-gw-1" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying a series of configuration commands and their outputs. The commands include enabling the router, configuring the terminal, setting up interfaces FastEthernet 0/1 and 0/0, and configuring the router's name and memory. The output shows the router's status and the successful configuration of the interfaces.

```
msk-donskaya-arromichina-gw-1
Physical Config CLI Attributes

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CTRL/Z.
Router(config)#interface FastEthernet 0/1
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface FastEthernet 0/0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#exit
Router#
AUI2-0-CONF1_1: Configured from console by console
enable
Router#enable
Translating "enable"...domain server (255.255.255.255)
A Unknown command or computer name, or unable to find computer address

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CTRL/Z.
Router(config)#no interface FastEthernet 0/1.2
Invalid interface type and number
Router(config)#end
Router#
AUI2-0-CONF1_1: Configured from console by console
write memory
Building configuration...
[OK]
Router#
```

Рис. 23: Настройка msk-donskaya-arromichina-gw-1

# Настройка file-arromichina 172.16.0.3

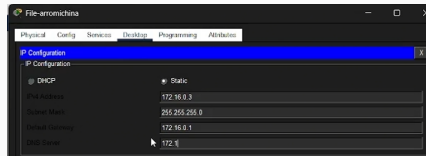


Рис. 24: Настройка file-arromichina 172.16.0.3

# Настройка other1-arromichina 172.16.0.3 аналогично настраиваем все нужные устройства

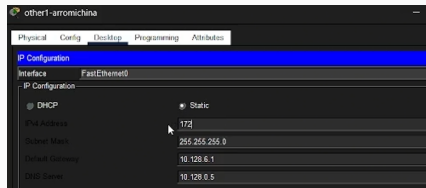


Рис. 25: Настройка other1-arromichina 172.16.0.3

# Настройка всех устройств на адрес 192.168.0.0

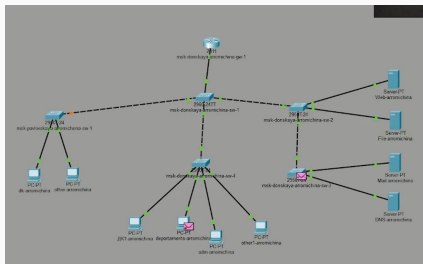


Рис. 26: Настраиваем все устройства на новый адрес 192.168.0.0

# Запускаем на проверку на этом этапе

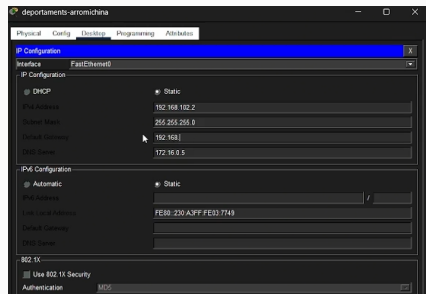


Рис. 27: Запускаем на проверку

Мы смогли познакомиться с принципами планирования локальной сети организации